

* Urinary System:

- Excretion: Removal of nitrogenous waste from body like urea.

* Excretion is the Elimination of metabolic waste materials like ammonia, urea, uric acid from the body.

- There are 3 Types of Excretion: Those are as follows.

1. Ammonotelism: Excretion of Ammonia from Ammonotelic organism.

* Ammonotelic organisms are those organism who excrete in water.

* Ammonia is very toxic material.

* Ammonia dissolves in water easily.

2. Uricotelic organisms are Aves (Birds)

* Excretion of uric acid from uricotelic organism is uricotelism.

3. Ureotelic organisms are Human.

* Excretion of urea from ureotelic organisms is ureotelism, i.e. Nitrogenous waste of body.

* Urine is less toxic.

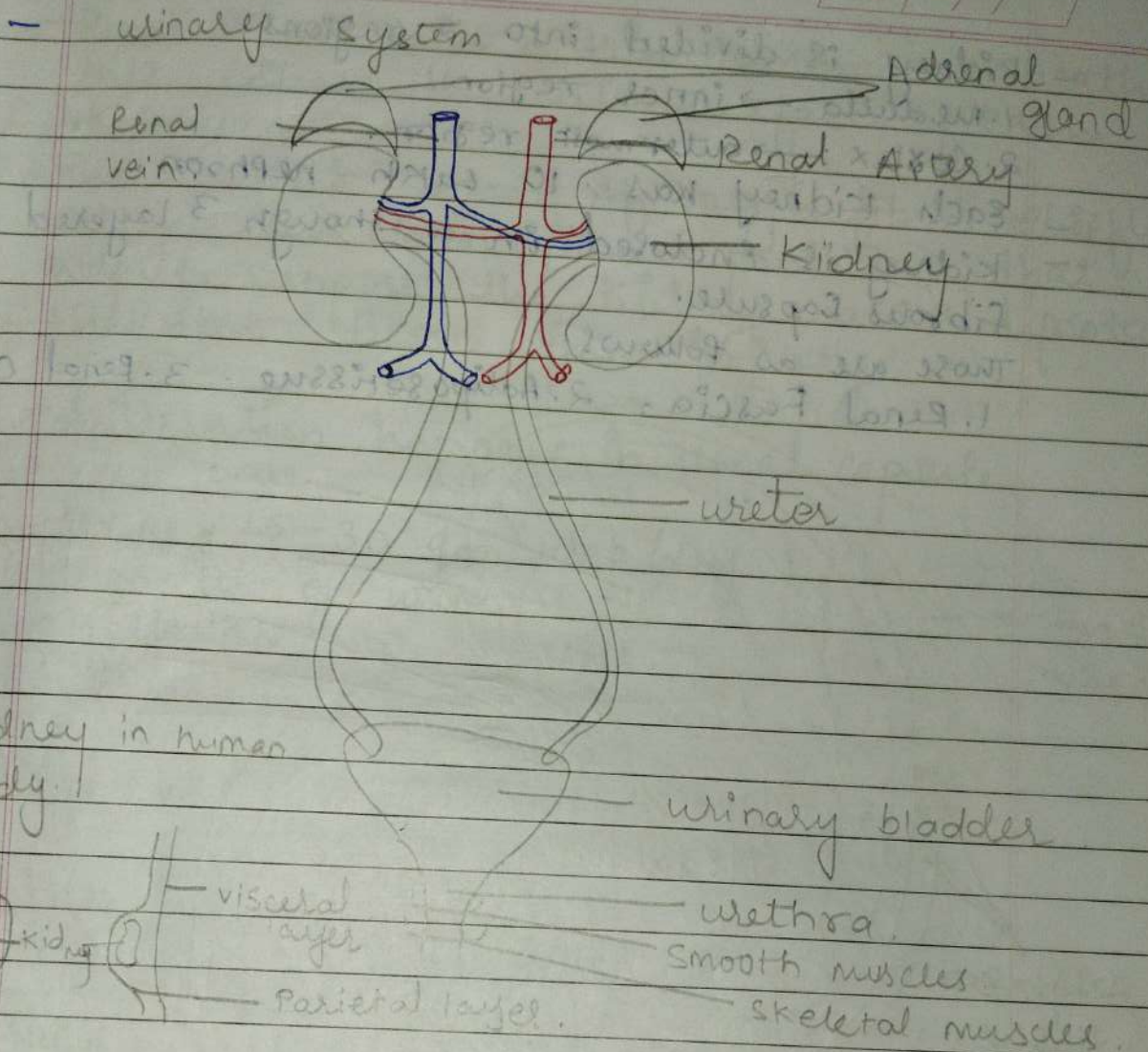
- Urinary System majorly consists of pair of kidney, pair of ureter, urinary bladder and urethra.

- Urinary bladder has smooth muscle and skeletal muscle.

- Aorta is largest Artery.

- Vena Cava is ^{largest} vein.

- Urea cycle takes place in liver.



* Kidney in human body.

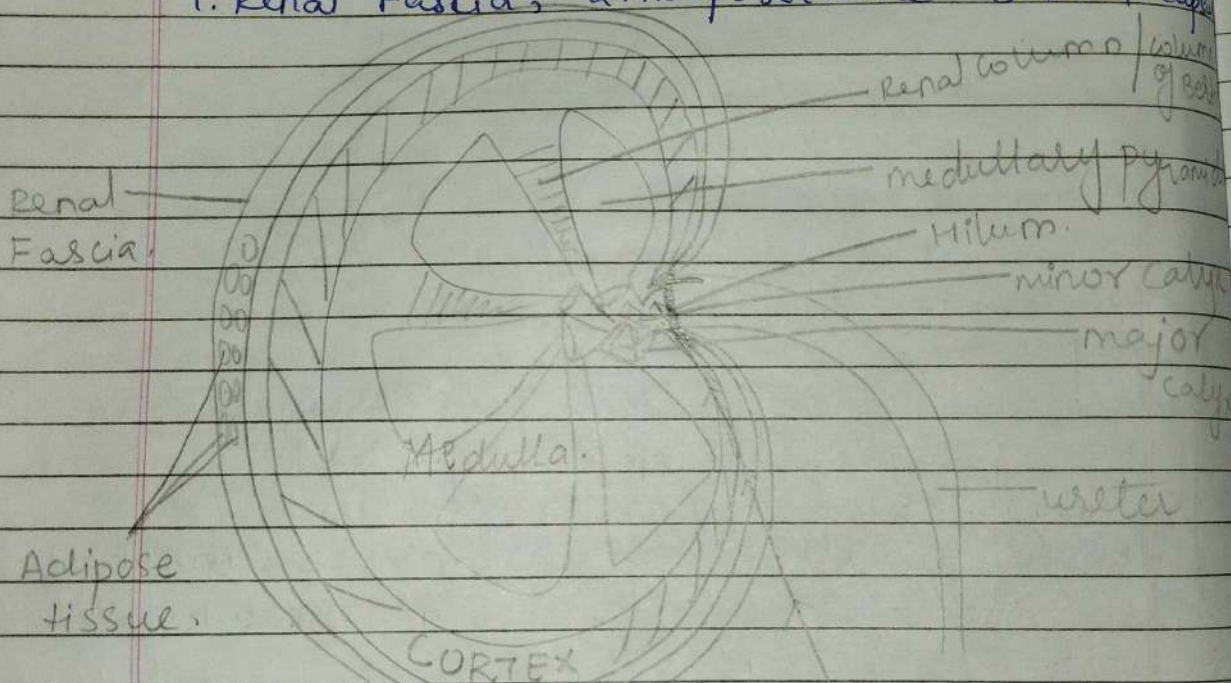
- * Kidney: - It is bean shaped organ.
- It is reddish brown in colour.
- Situated between the levels of last thoracic and 3rd Lumbar vertebra.
- 10-12 cm long in length.
- its width is 5-7 cm.
- It is a retroperitoneal organ.
- It is 2-3 cm thick.
- Average weight is 120-170 gm.
- ⊕ Renal vein (vena cava) have highest conc. of urea.
- ⊕ also called hepatic vein.

Kidney → Renal.

- Kidney is divided into 2 regions.
 1. Medulla → inner region.
 2. Cortex → outermost region.
- Each Kidney has 10 Lakh nephron.
- Kidney is Enclosed in a tough 3 layered fibrous Capsule.

Those are as follows.

1. Renal Fascia,
2. Adipose tissue.
3. Renal Capsule



- Renal Fascia anchors the Kidneys in place & (support) protects them from trauma and injury.
- Adipose Tissue provide Energy storage, insulation from extreme temperatures and cushioning around soft organ.

- Renal Capsule - Its a thin membranous sheath that covers the outer surface of each kidney.

- The capsule is composed of tough fibres, chiefly collagen and elastin (fibrous proteins), that help to support the kidney mass and protect the vital tissue from injury.

* Filtration happens in renal capsule.

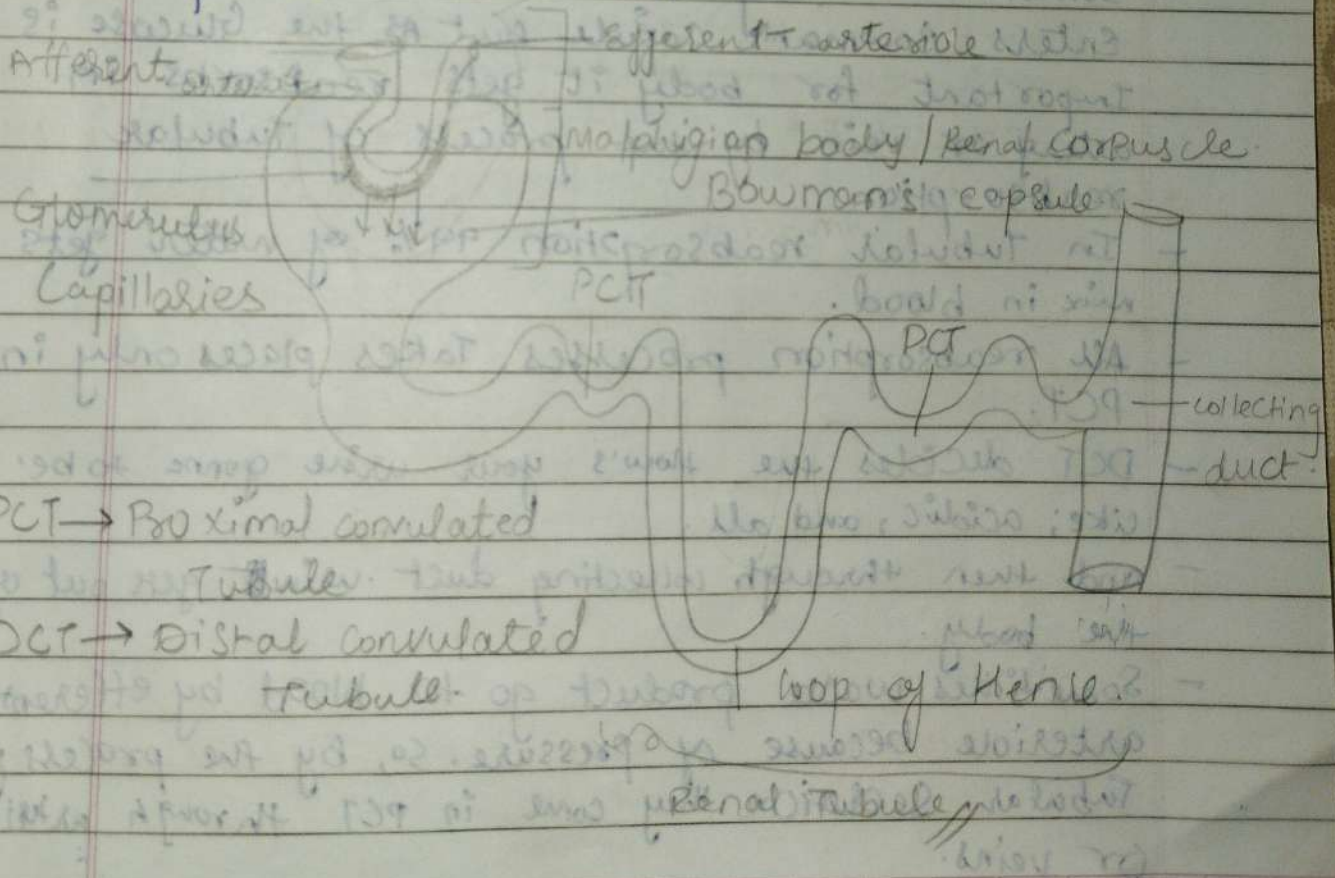
- Urine: • 25-30 gm urea/day.

• 1-1.5 ltr of urine.

• yellowish to colourless.

• urine PH $\rightarrow$ 6, i.e. acidic

* Nephron:



- It - ^{majorly} nephron consist of Malphigian body and renal Tubules.
- Malphigian body consists of Bowman's capsule which is also called as renal capsule, and Glomerulus Apparatus.
 - Glomerulus Apparatus consists of Afferent arteriole, Efferent arteriole and Glomerulus Capillaries.
 - Renal Tubules consist of PCT, loop of henle, DCT and CD.
 - Glomerulus capillaries are full of pores.
 - From the pores ions, waters and urea comes under Bowman's capsule by the process of Glomerular filtration.
 - Sometimes during Glomerular filtration Glucose enters Bowman's capsule but as the Glucose is important for body it gets reabsorbs in artery or vein by the process of Tubular reabsorption.
 - In Tubular reabsorption 99% of water gets mix in blood.
 - All reabsorption processes Takes places only in PCT.
 - DCT decides the how's your urine gonna to be like; acidic, and all.
 - and then through collecting duct urine gets out of the body.
 - Sometimes waste product go to blood by efferent arteriole because of pressure. So, by the process of Tubular Secretion they come in PCT through arteries or veins.

* Glomerular filtration.

→ filtration membrane: ① Squamous epithelium / Fenestra.

② Basement membrane.

③ Podocytes → Slit pores.

• Glomerular filtration rate (GFR).

- 180 L/day. → 125 ml/min.

- 1.5 L/day urine.

